# Understanding projectiles



AS Level | 9702 AS Ch.2 | Chapter 2: Accelerated motion | Lesson 2.14 | Sections 2.14

**Goal:** Solve problems for a projectile launched at an angle by resolving its initial velocity.

velocity

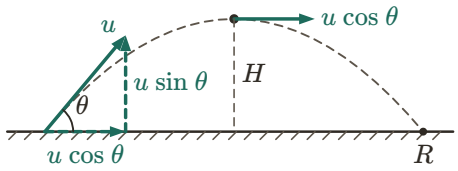
## 01 The big picture

Resolve the launch velocity into $u\cos\theta$ horizontally (constant all flight) and $u\sin\theta$ vertically (decelerated by $g$ on the way up). At the highest point the vertical velocity is zero, but the horizontal velocity is not.

Over level ground the flight is symmetrical: time up = time down. Without air resistance the range is greatest at 45°. Air resistance lowers the height and the range and makes the descent steeper.

**Range** [paraphrase]
The horizontal distance travelled before landing.

**Fig. 1 · Launched at an angle**




At the top only $u\cos\theta$ is left: the vertical velocity is zero.

## 02 Key equations

**Components**

$$u_x = u\cos\theta, \quad u_y = u\sin\theta$$

**Level ground**

$$T = \frac{2u_y}{g}, \quad R = u_x T$$

**Max height**

$$H = \frac{u_y^2}{2g}$$

**Symbols and units**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $u$, $\theta$ | launch speed and angle | $\mathrm{m\,s^{-1}}$, ° | $T$ | time of flight | s | $H$, $R$ | max height, range | m |

## 03 Sample problems with solutions

**Sample 1 · A kicked ball**

A ball is kicked at 20 m s$^{-1}$ at 30° above level ground. Find its time of flight, range and maximum height.

**Step 1.** Components:
$$u_x = 20\cos 30^\circ = 17.3\ \mathrm{m\,s^{-1}}, \quad u_y = 20\sin 30^\circ = 10.0\ \mathrm{m\,s^{-1}}$$

**Step 2.** Vertical gives the time; horizontal the range:
$$T = \frac{2\times 10.0}{9.81} = 2.04\ \mathrm{s}, \quad R = 17.3\times 2.04 = 35.3\ \mathrm{m}$$

**Step 3.** At the top $v_y = 0$:
$$H = \frac{10.0^2}{2\times 9.81} = 5.10\ \mathrm{m}$$

**Answer:** 2.0 s; 35 m; 5.1 m

Unit check: $\frac{(\mathrm{m\,s^{-1}})^2}{\mathrm{m\,s^{-2}}} = \mathrm{m}$ ✓

## 04 Practice problems

**P1** What is the velocity of a projectile at its highest point?
Answer: $u\cos\theta$, horizontal

**P2** Thrown at 25 m s$^{-1}$ at 60°: find the maximum height.
Answer: 23.9 m (24 m)

**P3** At 15 m s$^{-1}$ and 40°: find the time of flight.
Answer: 1.97 s (2.0 s)

**P4** Which launch angle gives the longest range?
Answer: 45° (no air resistance)

## 05 Other useful information

**Common mistakes**

✗ **Wrong:** Use $u$ in the vertical equations.
✓ **Right:** Vertical: $u\sin\theta$. Horizontal: $u\cos\theta$.

**Exam tips**

- Find $T$ from the vertical motion first; it links the two directions.

**What the syllabus asks**

- 9702 topic 2: projectile motion without air resistance. Coursebook 2.14.

**Explore:** Practice questions for this lesson